Introduction to Biophysics
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Solutions to Exercise Sheet 8

Problem 8.1

- ☐ True. The human karyotype comprises 22 autosomes and the two sex chromosomes, X and Y. Females have 22 autosomes and two X chromosomes for a total of 23 different chromosomes. Males also have 22 autosomes, but have an X and a Y chromosome for a total of 24 different chromosomes.
- ☐ True. Humans and mice diverged from a common ancestor long enough ago for roughly 2 out of 3 nucleotides to have been changed by random mutation. The regions that have been conserved are those with important functions, where mutations with deleterious effects were eliminated by natural selection. Other regions have not been conserved because natural selection cannot operate to eliminate changes in nonfunctional DNA.
- ☐ True. All the core histones are rich in lysine and arginine, which have basic-positively charged-side chains that can neutralize the negatively charged DNA backbone.
- ☐ False. By using the energy of ATP hydrolysis, chromatin remodeling complexes can catalyze the movement of nucleosomes along DNA, or even between segments of DNA.
- ☐ True. Duplication of chromosomal segments, which may include one or more genes, allows one of the two genes to diverge over time to acquire different, but related functions. The process of gene duplication and divergence is thought to have played a major role in the evolution of biological complexity.

Problem 8.2

A dicentric chromosome is unstable because the two kinetochores have the potential to interfere with one another. Normally, microtubules from the two poles of the spindle apparatus attach to opposite faces of a single kinetochore in order to separate the individual chromatids at mitosis. If a chromosome contains two centromeres, half of the time the microtubules from one of the poles will attach to the two kinetochores associated with one chromatid, while the microtubules from the other pole will attach to the two kinetochores associated with the other chromatid. Division can then occur satisfactorily. The other half of the time, the microtubules from each pole will attach to kinetochores that are associated with different chromatids. When that happens, each chromatid will be pulled to opposite spindle poles with enough force to snap it in two. Thus, two centromeres are bad for a chromosome, causing chromosome breaks-rendering it unstable.

Problem 8.3

With these assumptions, the DNA is compacted 27-fold in 30-nm fibers relative to the extended DNA. The total length of duplex DNA in 50 nm of the fiber is 1360 nm [(20 nucleosome) $\times$ (200 bp/nucleosome) $\times$ (0.34 nm/bp) = 1360 nm). 1360 nm of duplex DNA reduced to 50 nm of chromatin fiber represents a 27-fold condensation [(1360 nm/50 nm) = 27.2]. This level of packing represents 0.27% (27/10000) of the total condensation that occurs at mitosis, still a long way off what is needed.